

```
import math

n = int(input("Enter n: "))

log_value = math.log2(n)
result = n * log_value

print("Recurrence:  $T(n) = 2T(n/2) + n$ ")
print("a = 2")
print("b = 2")
print("f(n) = n")
print("Time Complexity =  $\Theta(n \log n)$ ")
print("For n =", n, "n log2(n) =", result)
```

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [
on win32
Enter "help" below or click "Help" above for more information

>>>

==== RESTART: C:/Users/ROOPA SAI DIVYA/OneDrive/Documents/Big
Enter n: 5
Recurrence:  $T(n) = 2T(n/2) + n$ 
a = 2
b = 2
f(n) = n
Time Complexity =  $\Theta(n \log n)$ 
For n = 5 n log2(n) = 11.60964047443681
```

>>>